

the best clock in the network to become the master—typically a GPS signal but could be any clock. If the master clock ‘dies’ then the next best clock in the network is selected as master. For automotive networks, it is likely that the BMCA will not be used and the master/slave configuration of each node will be fixed.

What are the major benefits of using standard Ethernet in automotive systems?

Standard Ethernet can provide complete network ubiquity across automotive applications. One of the major benefits of standard Ethernet is the economies of scale across mass markets deploying Ethernet, providing lowered total cost of ownership. Standard Ethernet devices can be used in automotive, that is, the same silicon used for automotive Ethernet devices will also be utilised in all other Ethernet markets, such as, enterprise, telecom, digital home and industrial.

What key requirements exist for automotive applications?

One of the key requirements for automotive applications when the car is moving (unlike diagnostics in a garage) is the need to meet OEM EMI limits over unshielded cable, which is preferred by automotive OEMs to reduce cable costs and ease installation.

How can an engineer improve the data security of an automotive Ethernet-enabled system?

Micrel being a silicon vendor focusses primarily on this from the low level (physical/device layer). There are various security features within Ethernet, such as, virtual LANs that can be implemented within the same physical network by tagging packets. Here we can create various virtual networks within one physical network. For instance, in diagnostics, even though it is physically connected to the rest of the car, when we plug in to the diagnostics, these would be blocked from any forbidden access and only certain devices can be accessed.

Can we compare standard Ethernet with CAN and Flexray?

Today we cannot compare Ethernet directly with CAN. It will be a number of years before Ethernet is considered to replace CAN interfaces. CAN, whilst it is cheap, can only support low bandwidths, typically less than 1Mbps.

What alternatives exist for standard Ethernet?

For infotainment networking, one of the alternative technologies is MOST, originally offered by SMSC, now acquired by Microchip. The major drawback with MOST technology is that, although it is an open standard that can be licenced, there is only one major supplier and it is limited to automotive applications only. One of the concerns with such reversed engineered vendor standards is the interoperability.

MOST is also limited in the available bandwidth due to speed and network topology. Here a maximum of 150Mbps must be shared across all network nodes in the necessary ring topology, whereas Ethernet can provide 100Mbps, 1Gbps or even greater per node, irrespective of network size.

What are the top reasons for an engineer to switch to standard Ethernet?

In any application, irrespective of market, standard Ethernet provides the lowest total cost of ownership for high-bandwidth communications.

The modern car of today may comprise one hundred or more microprocessors, each requiring programming. Using traditional method over a CAN bus will take hours to re-flash such a car, resulting in a major cost to the car manufacturer. The increased bandwidth provided by Ethernet reduces this time to mere minutes and can be ‘re-programmed whilst you wait’—connected to a standard PC or laptop.

The key to the success of standard Ethernet and the benefits it brings to any network is the true open standardisation of IEEE 802.3.

What differs between standard Ethernet and the competing BroadR-Reach?

BroadR-Reach technology, developed by Broadcom, was a proprietary PHY layer device designed for extending the reach of 100Mbps Ethernet data. Originally, it was perceived by the industry that it was not possible to meet automotive emission limits utilising standard Ethernet over unshielded twisted cable, which led to the interest in alternative possible technologies, such as BroadR-Reach. However, Micrel and other suppliers have since demonstrated to the industry this to be achievable with standard Ethernet. Unshielded cable is highly desirable for car manufactures to reduce cost and ease installation.

BroadR-Reach uses additional signal processing, similar to gigabit technology, although not interoperable consuming additional power. In addition, only standard Ethernet can support other IEEE standards such as IEEE802.3az energy-efficient Ethernet, whereby, during idle periods, the transceivers can fall back into a low-power sleep mode reducing power consumption by an additional 50 per cent or more. Standard Ethernet provides another major benefit with support of IEEE802.af/at power over Ethernet. Here a remote device, for example rear-view camera, can be supplied power over the same cable as you send data.

Like our other comparisons, the major benefit of standard Ethernet compared to BroadR-Reach or any of the other alternatives is the true open standardisation of IEEE 802.3. The IEEE 802.3 Ethernet eco-system is mature, and proven, providing the design community with design, test and conformance specifications and a wealth of off-the-shelf test equipment solutions. ●

Dilip Anand is a senior assistant editor at EFY, Bengaluru, and Sneha Ambastha is a technical correspondent at EFY, Gurgaon